

Animal Nutrition

Science

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Short Title: Animal Nutrition
Faculty Science and Computing
Department Science
Cluster Agriculture
Author SWALSH
Credits 5

Level: Introductory

Description of Module / Aims

This module will give students a broad introduction to both ruminant and monogastric nutrition for farm animals. Students will learn about the basic principles and key concepts of nutrition, and its importance in diet formulation and prevention of metabolic diseases and disorders.

Programmes

AGRI -0105 BSc (Hons) in Agricultural Science (WD_SAGRI_B)

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Indicative Content

- Structure and function of ruminant and monogastric digestive systems
- Metabolism of proteins, carbohydrates, fatty acids and amino acids
- Nutrient requirements and intake: macro- and micronutrients
- Systems of energy and protein evaluation of feeds
- Feeding standards for maintenance, growth, reproduction and lactation
- Nutrient sources: grass, conserved forages, root crops, cereals & processed forage
- Diet and ration formulation
- Metabolic diseases and disorders

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain digestion and metabolism in ruminants and monogastrics.
2. Identify the nutrient requirements and intake potential of ruminants in terms of energy, protein, fibre and minerals.
3. Identify a range of feed and forage sources.
4. Calculate diet and ration formulations feeding regimes for monogastric & ruminant production systems.
5. Explain the effect of nutrition on metabolic diseases and disorders for dairy, beef, sheep, pigs & poultry.
6. Use a range of laboratory techniques to classify and calculate the nutritional value of feedstuffs for ruminants.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Guest industry speakers.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,4,5
Continuous Assessment	50%	
<i>Lab Report</i>	50%	3,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter.

40%-49%: Will be able to show a good knowledge of the key learning outcomes including anatomy and functions of ruminant and monogastric digestive systems, as well as a basic knowledge of macro- and micro-nutrients.

50%-59%: As well as the above, the student will be able to demonstrate a good understanding of the learning outcomes as well as being able to identify and discuss various nutrient sources.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics including how to predict nutritional requirements of the animal and potential nutritional disorders.

70%-100%: The student will be able to demonstrate a comprehensive knowledge and understanding of all learning outcomes, including the ability to fully integrate the various components of the course.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Chahal, U.S., P.S. Niranjana and S. Kumar. *Handbook of general animal nutrition*. UK: IBDC, 2008.

Supplementary Material

- D'Mello, J.P.F. *Farm Animal Metabolism and Nutrition*. UK: CABI Publishing, 2000.
- Forbes, J.M. *Voluntary Food Intake and Diet Selection in Farm Animals*. UK: CABI Publishing, 2007.
- McDonald, P., R.A. Edwards, J.F. Greenhalgh and C.A. Morgan. *Animal Nutrition*. UK: Pearson, 2002.
- McDowell, L.R. *Vitamins in Animal and Human Nutrition*. UK: Wiley-Blackwell, 2008.

Requested Resources

- Room Type: Biology Lab